Transit-Based Analysis of Three Candidate Exoplanet Systems

***Abstract*—This report analyzes the brightness observations of STAC-1, STAC-2, and STAC-3 to investigate possible transiting exoplanets. Repeated brightness dips were used to estimate orbital periods, and the supplied transit-depth and Kepler's Third Law relations were applied to estimate planet radii and orbital distances. The estimated periods are 4, 80, and 40 days, respectively. Among the three candidates, STAC-3's estimated orbit is closest to its approximate habitable-zone distance.**

I. INTRODUCTION

A planet passing between an observer and its host star can produce a small, recurring decrease in measured brightness. This transit method motivates the analysis. The task is to inspect the observations for possible periodic dips, estimate candidate planet properties, and identify one candidate for further study. The results below are estimates based on the provided data of the time v/s relative brightness.

II. DATA AND METHOD

The stellar masses, radii, and luminosities are taken from the problem statement. Relative brightness values below 1.0000 are treated as possible transit observations. The adopted candidate dip days are STAC-1: 3, 8, 12, 16, and 20; STAC-2: 20, 100, and 180; and STAC-3: 20, 60, 100, and 140. The typical gaps suggest periods of approximately 4, 80, and 40 days, respectively. The first STAC-1 gap is 5 days rather than 4, and STAC-3 has another low measurement at day 170 that does not fit the adopted pattern.

The transit depth δ represents the relative decrease in stellar brightness during a planetary transit and is defined as:

$$\delta = (F_{normal} - F_{transit}) / F_{normal} \quad (1)$$

where F_{normal} is the baseline flux (normalized stellar brightness out-of-transit, set to 1.0000), $F_{transit}$ is the measured flux during transit (in-transit brightness), and δ is the fractional depth.

The planet radius R_p is derived from the star-planet area ratio during transit:

$$R_p = R_s \sqrt{\delta} \quad (2)$$

where R_s is the host star radius (in solar radii $R_\odot$) and R_p is the planetary radius (in kilometers or Earth radii $R_\oplus$).

The semi-major axis orbital distance a is calculated using Kepler's Third Law rearranged for circular orbits:

$$a = (M P^2)^{1/3} \quad (3)$$

where P is the planet's orbital period in Earth years (yr), M is the host star mass in solar masses ($M_\odot$), and a is the semi-major axis in Astronomical Units (AU).

The stellar habitable-zone center distance d_{HZ} is derived using the luminosity relation:

$$d_{HZ} = \sqrt{(L / L_\odot)}\ AU \quad (4)$$

where L is the stellar luminosity, $L_\odot$ is solar luminosity, and d_{HZ} is in AU.

III. RESULTS

A. DETECT THE EXOPLANETS

STAC-1 System

Recorded transit dip days are t = 3, 8, 12, 16, and 20. Disregarding the initial 5-day gap, recurring intervals yield $P_1 \approx$ 4 days (4/365.25 = 0.01095 yr). In-transit brightness dips to $F_{transit}$ = 0.9970 from F_{normal} = 1.0000, yielding transit depth δ_1 = (1.0000 − 0.9970)/1.0000 = 0.0030 (0.30%).

STAC-2 System

Recorded transit dip days are t = 20, 100, and 180. The uniform 80-day interval yields P_2 = 80 days (80/365.25 = 0.2190 yr). In-transit brightness drops to $F_{transit}$ = 0.9990 from F_{normal} = 1.0000, yielding transit depth δ_2 = (1.0000 − 0.9990)/1.0000 = 0.0010 (0.10%).

STAC-3 System

Recorded major transit dip days are t = 20, 60, 100, and 140 (ignoring anomalous low reading at day 170). The recurring 40-day interval yields P_3 = 40 days (40/365.25 = 0.1095 yr). Minimum brightness drops to $F_{transit}$ = 0.9990 from F_{normal} = 1.0000, yielding transit depth δ_3 = (1.0000 − 0.9990)/1.0000 = 0.0010 (0.10%).

A periodic decrease in stellar brightness indicates an exoplanet because an orbiting planet periodically passes between its host star and the observer, blocking a portion of starlight proportional to its cross-sectional area.

B. DETERMINE THE PLANET SIZE

Using host star radii R_s from the problem statement, $R_\odot$ = 696,000 km, and $R_\oplus$ = 6,371 km [1]:

STAC-1 System

Given $R_{s,1}$ = 1.00 $R_\odot$ and δ_1 = 0.0030:

$$R_{p,1} = 1.00(696{,}000\ km)\ \sqrt{0.0030} \approx 38{,}121\ km \approx 5.98\ R_\oplus \quad (5)$$

Classification: **Neptune-sized**.

STAC-2 System

Given $R_{s,2}$ = 0.80 $R_\odot$ and δ_2 = 0.0010:

$$R_{p,2} = 0.80(696{,}000\ km)\ \sqrt{0.0010} \approx 17{,}607\ km \approx 2.76\ R_\oplus \quad (6)$$

Classification: **Super-Earth / Sub-Neptune**.

STAC-3 System

Given $R_{s,3}$ = 0.50 $R_\odot$ and δ_3 = 0.0010:

$$R_{p,3} = 0.50(696{,}000\ km)\ \sqrt{0.0010} \approx 11{,}005\ km \approx 1.73\ R_\oplus \quad (7)$$

Classification: **Super-Earth**.

C. DETERMINE THE PLANETARY ORBIT

Substituting period P in years and stellar mass M in solar masses $M_\odot$ into Kepler's Third Law ($a = [M P^2]^{1/3}$), converted to kilometers using 1 AU = 149.6 million km and compared step-by-step to Earth's distance from the Sun (1.00 AU = 149.6 million km):

STAC-1 System

Given $M_1 = 1.00\ M_\odot$ and $P_1 = 0.01095$ yr:

$$a_1 = [1.00\ (0.01095)^2]^{1/3} \approx 0.049\ AU \quad (8)$$

Kilometer Conversion: $a_1 = 0.049 \times 149.6$ million km ≈ 7.33 million km.

Earth Comparison Ratio: $a_1 / a_{Earth} = 0.049$ AU / 1.00 AU = 0.049 (20.4 times closer than Earth to the Sun).

STAC-2 System

Given $M_2 = 0.80\ M_\odot$ and $P_2 = 0.2190$ yr:

$$a_2 = [0.80\ (0.2190)^2]^{1/3} \approx 0.337\ AU \quad (9)$$

Kilometer Conversion: $a_2 = 0.337 \times 149.6$ million km ≈ 50.42 million km.

Earth Comparison Ratio: $a_2 / a_{Earth} = 0.337$ AU / 1.00 AU = 0.337 (2.97 times closer than Earth to the Sun).

STAC-3 System

Given $M_3 = 0.50\ M_\odot$ and $P_3 = 0.1095$ yr:

$$a_3 = [0.50\ (0.1095)^2]^{1/3} \approx 0.182\ AU \quad (10)$$

Kilometer Conversion: $a_3 = 0.182 \times 149.6$ million km ≈ 27.23 million km.

Earth Comparison Ratio: $a_3 / a_{Earth} = 0.182$ AU / 1.00 AU = 0.182 (5.49 times closer than Earth to the Sun).

D. FIND THE MOST PROMISING WORLD

Using $d_{HZ} = \sqrt{(L / L_\odot)}$ AU: $d_{HZ,1} = \sqrt{1.00} = 1.000$ AU; $d_{HZ,2} = \sqrt{0.40} \approx 0.632$ AU; $d_{HZ,3} = \sqrt{0.08} \approx 0.283$ AU.

Candidate Selection: STAC-3 is determined to be the most promising candidate for further habitability studies.

Supporting Evidence:

• *Habitable Zone Proximity:* STAC-3's orbital distance (0.182 AU) is closest to its habitable zone center (0.283 AU), with an offset of only 0.101 AU. STAC-1 (0.049 AU vs 1.000 AU) orbits at 0.049 AU around a 1.00 $L_\odot$ star, receiving over 400 times Earth's stellar flux and rendering it extremely hot. STAC-2 (0.337 AU vs 0.632 AU) is also significantly offset.

• *Terrestrial Size:* With an estimated radius of 1.73 $R_\oplus$ (Super-Earth), STAC-3 is small enough to possess a rocky surface, unlike STAC-1 (5.98 $R_\oplus$, Neptune-sized gas planet) and STAC-2 (2.76 $R_\oplus$, sub-Neptune).

• *Thermal Balance:* STAC-3's low stellar luminosity allows a close orbital distance of 0.182 AU to maintain moderate surface temperatures compatible with liquid water.

Additional Measurement Suggestion: Transit transmission spectroscopy should be performed to detect atmospheric composition and key biosignatures such as water vapor, carbon dioxide, or methane.

TABLE I
APPROXIMATE PROPERTIES OF CANDIDATE PLANETS

System	P (d)	δ	$R_p / R_\oplus$	a (AU)	d_{HZ} (AU)
STAC-1	4	0.0030	5.98	0.049	1.000
STAC-2	80	0.0010	2.76	0.337	0.632
STAC-3	40	0.0010	1.73	0.182	0.283

IV. CANDIDATE COMPARISON

STAC-3 is selected as the most promising candidate for further study in this comparison. Its estimated orbital distance, 0.182 AU, is closest to its approximate habitable-zone distance, 0.283 AU, among the three candidates. Its estimated radius, 1.73 Earth radii, is also the smallest of the three. These observations do not confirm an atmosphere, liquid water, or life, as the habitable-zone relation is a preliminary distance estimate.

V. LIMITATIONS AND FURTHER OBSERVATIONS

The time sampling is sparse, so transits could occur between observations. STAC-1's first candidate dip is not consistent with the later four-day spacing, and STAC-3 has an extra dip at day 170 that does not match the 40-day sequence. A dip could also result from measurement noise or stellar variability. More frequent brightness observations over several additional cycles could test whether dips recur reliably.

VI. CONCLUSION

The observations contain repeated brightness dips consistent with transiting planets. The adopted periods are approximately 4 days for STAC-1, 80 days for STAC-2, and 40 days for STAC-3. Calculations yield radii of 5.98, 2.76, and 1.73 Earth radii and orbital distances of 0.049, 0.337, and 0.182 AU. STAC-3 is prioritized for additional observations because its orbit is closest to its habitable zone.

REFERENCES

1. STAC WARS, "PS 3: Exoplanet Hunters Operation Dim Star," hackathon problem statement, pp. 1–5.
2. Google Gemini, document formatting and report structure according to IEEE guidelines.
3. Wikipedia.
4. Google.